

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20 Total Marks:20 Annexure A

Mock Interview

Code of Conduct:

I Cheredla Jasvina/192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Annexure A

Mock Interview

1. What is Eigenfaces?

Ans :

Eigenfaces is a face-recognition technique based on PCA (Principal Component Analysis). A set of face images is used to calculate principal components called eigenfaces. Each face is represented as a combination of these eigenfaces, reducing dimensionality and making recognition faster.

2. Define PCA in image processing.

Ans :

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms correlated image features into a smaller set of uncorrelated variables called principal components. It preserves the maximum possible variance while reducing the number of features.

3. What is contour-based shape representation?

Ans :

Contour-based representation describes an object using its boundary or outline. Shape information is obtained from the pixels forming the object's contour. Examples include chain codes, polygonal approximation, Fourier descriptors, and curvature-based methods.

4. What is region-based shape representation?

Ans :

Region-based representation describes a shape using the entire area inside its boundary, rather than only its contour. Features include area, centroid, moments, compactness, and texture. It is useful when the internal structure of an object is important.

5. How does an active contour (snake) work?

Ans :

An active contour or snake is a deformable curve that moves toward an object's boundary. It **minimizes an energy function containing:**

Internal energy – controls smoothness and elasticity.

External/image energy – attracts the curve toward edges or other image features.

Constraint energy – incorporates user or application-specific information.

The curve iteratively changes until it fits the desired object boundary.

6. How does a level set method work?

Ans :

A level set method represents a contour implicitly as the zero level of a higher-dimensional function. Instead of directly moving the contour, the function is evolved according to image information. This allows the contour to split, merge, and handle complex shapes naturally.

7. What is the Hough Transform?

Ans :

The Hough Transform is a feature-detection technique used to identify geometric shapes such as lines, circles, and ellipses. Image points vote in a parameter space, and peaks in that space indicate the presence of the corresponding shapes.

8. How are Fourier descriptors used in shape analysis?

Ans :

Fourier descriptors represent an object's contour using its Fourier transform. The contour coordinates are converted into frequency components. A small number of dominant coefficients can describe the overall shape, providing compact and often noise-resistant shape features.

9. How are wavelet descriptors used in shape analysis?

Ans :

Wavelet descriptors represent shape contours using wavelet transforms. Wavelets analyze the contour at different scales and frequencies, capturing both global and local shape characteristics. They are useful for detecting fine details while maintaining robustness to noise.

10. What is a particle filter?

Ans :

A particle filter is a probabilistic tracking method that represents the possible state of an object using a set of weighted samples called particles. It repeatedly:

- Predicts possible new states.

- Compares them with observations.

- Assigns weights according to their likelihood.

- Resamples the particles.

It is particularly useful for nonlinear and non-Gaussian tracking problems.

11. How is foreground–background separation done in surveillance systems?

Ans :

Foreground–background separation identifies moving objects by distinguishing them from the **stationary scene. Common methods include:**

- Background subtraction

- Frame differencing

- Gaussian Mixture Models (GMM)

- Adaptive background models

- Deep-learning-based segmentation

The resulting foreground regions can then be used for person detection and tracking.

12. How do multi-camera systems track pedestrians?

Ans :

Multi-camera pedestrian tracking combines observations from multiple cameras. The system:

- Detects pedestrians in each camera.

- Extracts appearance and motion features.

- Matches the same person across different camera views.

Estimates their position and movement.

Maintains a common identity across cameras.

This is called multi-camera or cross-camera tracking.

13. What is a Convolutional Neural Network (CNN)?

Ans :

A CNN is a deep-learning network designed primarily for processing images. It uses convolutional layers to automatically learn features such as edges, textures, shapes, and objects. Pooling and fully connected layers are then used for tasks such as classification, detection, and segmentation.

14. How does a Vision Transformer (ViT) work?

Ans :

A Vision Transformer (ViT) divides an image into small patches and converts each patch into an embedding. These embeddings are processed by a Transformer using self-attention, which learns relationships between different image regions. A classification head then produces the final prediction.

15. What is a Generative Adversarial Network (GAN)?

Ans :

A GAN consists of two neural networks:

Generator – creates synthetic images.

Discriminator – tries to distinguish real images from generated ones.

The two networks compete during training, causing the generator to produce increasingly realistic images.

16. How is a Support Vector Machine (SVM) used for image classification?

Ans :

An SVM classifies images by finding a decision boundary, called a hyperplane, that maximizes the margin between different classes. Image features such as HOG, SIFT, color features, or CNN features can be supplied to the SVM. Kernels can also be used to handle nonlinear classification.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

Ans :

An HMM models an activity as a sequence of hidden states that produce observable features. For example, walking may consist of sequential states such as different phases of movement. Motion or pose features are given to the HMM, which estimates the most likely sequence of activities.

18. What is a Gaussian Mixture Model (GMM)?

Ans :

A Gaussian Mixture Model represents data as a weighted combination of several Gaussian probability distributions. Each Gaussian represents a cluster or component. In image processing, GMMs are commonly used for background modeling, segmentation, clustering, and object detection.

19. How is graph-based learning applied to 3D shape correspondence?

Ans :

In graph-based learning, 3D shapes are represented as graphs, where vertices represent points or regions and edges represent relationships between them. A graph-learning model learns similarities between the graphs and establishes corresponding points or regions. This is useful for 3D object matching, registration, and shape analysis.

20. How is road and lane detection performed in autonomous driving?

Ans :

Road and lane detection generally involves:

Image acquisition using cameras or other sensors.

Preprocessing to reduce noise and improve visibility.

Road segmentation to identify drivable areas.

Lane detection using methods such as edge detection, Hough Transform, polynomial fitting, or deep neural networks.

Lane tracking across successive frames.

Combining the results with other sensors such as LiDAR, radar, and GPS for reliable vehicle navigation.

Modern autonomous-driving systems increasingly use CNNs, Transformers, and semantic/instance segmentation networks for robust road and lane detection.